

KIVALINA, AK, USA

WMO: 701486

Lat: 67.732N

Lon: 164.548W

Elev: 10

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 99-19

WBAN: 26642

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.5	-28.3	-33.6	0.8	-26.1	-30.7	1.0	-24.1	40.5	1.0	36.7	2.0	4.7	70	0.980

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	65.3	57.2	61.8	55.3	58.5	53.7	58.7	64.1	56.5	60.7	54.3	57.6	10.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.7	66.1	60.7	54.1	62.2	58.8	52.2	58.1	56.7	25.5	64.7	24.1	60.4	22.8	57.7	67.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	35.0	31.5	27.2	DB	-37.7	71.2	8.0	4.5	-43.5	74.5	-48.2	77.2	-52.7	79.7	-58.5	83.0
(5)				WB	-32.1	60.7	3.8	2.4	-34.9	62.4	-37.1	63.8	-39.3	65.1	-42.0	66.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.2	0.2	2.8	0.8	14.3	31.7	43.9	50.3	49.6	41.3	26.8	11.0	4.0
	DBStd	22.03	16.87	16.77	12.28	11.83	8.80	7.65	5.69	5.24	6.53	9.33	11.90	13.72
	HDD50	9962	1543	1321	1527	1070	568	216	64	71	266	719	1171	1426
	HDD65	15269	2008	1741	1992	1520	1031	636	457	478	710	1184	1621	1891
	CDD50	171	0	0	0	0	2	32	73	58	6	0	0	0
	CDD65	2	0	0	0	0	0	1	1	0	0	0	0	0
	CDH74	19	0	0	0	0	0	16	3	0	0	0	0	0
	CDH80	6	0	0	0	0	0	6	0	0	0	0	0	0
Wind	WSAvg	12.4	13.2	13.5	12.6	11.7	10.4	10.1	10.7	12.2	12.7	12.9	13.7	14.9
Precipitation	PrecAvg	17.8	1.3	1.4	1.0	0.8	0.7	0.8	1.9	2.8	2.0	2.4	1.5	1.2
	PrecMax	24.8	2.7	4.4	5.0	2.2	1.8	1.5	3.9	6.1	4.4	4.6	4.6	3.1
	PrecMin	12.8	0.3	0.2	0.2	0.1	0.0	0.4	0.5	0.6	0.5	0.9	0.2	0.4
	PrecStd	3.1	0.6	1.0	0.9	0.6	0.5	0.3	0.8	1.3	0.8	0.9	1.0	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.2	39.8	30.3	38.7	58.7	72.0	72.0	69.1	59.2	48.8	35.9	34.3
		MCWB	33.2	33.4	29.0	35.2	49.2	57.4	60.9	59.2	50.6	40.2	34.0	32.4
	2%	DB	32.4	34.4	28.0	36.4	50.5	63.1	65.5	64.1	55.1	43.4	33.4	32.0
		MCWB	30.5	31.0	26.6	33.9	44.7	53.3	58.3	57.3	49.5	40.0	32.3	30.8
	5%	DB	28.9	30.6	25.5	34.7	46.2	58.2	61.9	61.0	52.6	40.8	31.2	29.2
		MCWB	27.5	29.2	23.8	32.7	41.6	52.2	56.3	55.6	49.1	38.6	30.2	28.1
	10%	DB	25.3	27.2	19.9	32.7	41.9	54.5	58.4	57.5	50.7	38.4	28.8	25.1
		MCWB	23.6	25.7	18.5	31.1	38.6	50.2	54.5	53.4	47.8	36.4	27.7	23.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.1	34.5	29.7	35.5	49.4	59.2	62.1	61.1	53.0	42.8	34.6	33.3
		MCDB	35.8	37.8	30.1	37.7	56.6	68.1	69.9	66.2	56.1	45.7	35.4	33.8
	2%	WB	30.7	31.7	26.9	34.2	45.6	55.0	59.0	58.5	51.2	40.6	32.4	30.6
		MCDB	32.1	34.1	27.9	36.0	50.5	60.7	64.8	63.2	53.8	42.9	33.2	31.5
	5%	WB	27.9	29.4	24.1	33.0	41.7	52.6	57.0	56.2	49.8	38.9	30.2	27.9
		MCDB	29.2	31.0	25.7	34.5	45.8	57.5	61.3	60.0	52.3	41.0	31.0	28.9
	10%	WB	24.2	26.0	19.0	30.9	38.8	50.3	54.9	54.0	48.0	36.5	27.7	23.9
		MCDB	25.8	27.7	20.3	32.4	41.9	55.0	58.0	56.9	50.6	38.2	28.7	25.4
Mean Daily Temperature Range	5% DB	MDBR	13.6	14.6	15.0	15.1	10.7	10.5	9.3	9.9	9.9	8.9	10.2	11.8
		MCDBR	13.3	14.2	14.3	12.4	15.8	17.8	15.9	15.0	11.3	7.7	7.1	10.1
		MCWBR	12.5	12.6	13.8	10.9	11.0	10.8	10.0	9.7	7.8	6.3	7.1	9.7
	5% WB	MCDBR	12.9	14.1	13.9	11.9	14.9	16.5	14.0	12.9	9.8	7.0	6.8	10.7
		MCWBR	12.3	12.6	13.5	10.7	10.8	10.5	9.3	8.9	7.2	6.0	7.1	10.4
Clear-Sky Solar Irradiance	taub	0.171	0.235	0.258	0.322	0.361	0.346	0.355	0.363	0.306	0.253	N/A	N/A	
	taud	1.899	2.078	2.213	2.056	2.075	2.391	2.452	2.412	2.519	2.282	N/A	N/A	
	Ebn at Noon	46	184	253	258	260	272	265	248	239	180	N/A	0	
	Edh at Noon	3	15	24	38	42	32	29	27	19	12	N/A	0	
All-Sky Solar Radiation	RadAvg	15	173	725	1421	1828	1952	1517	1092	653	214	31	0	
	RadStd	2	22	52	71	157	164	158	137	67	27	4	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+2.57	N/A	N/A	+1.80	+1.48	N/A	-855	-918	+117	+2

Nomenclature: See separate page